

SAWYER CAWTHON

[CTI/scripts/triple_extraction at main · edogdu/CTI](#)

1. Objective

To improve the precision and contextual coherence of extracted CTI triples by integrating semantic chunking into the LLM-based triple-extraction and validation pipeline.

This addresses noise in sentence-level extraction and ensures that related entities within a single semantic topic are processed as a unit before ontology validation.

Stage	Description
a. Document Conversion	Convert PDF reports via <code>docling.DocumentConverter</code> , preserving provenance and page layout.
b. Sentence Segmentation	Apply <code>spaCy</code> sentencizer to extract sentences per page.
c. Semantic Chunking (Max-Min Method)	Embed sentences using <code>nomic-embed-text</code> via Ollama; group adjacent sentences by maximizing the minimum cosine similarity within each chunk (Kiss et al., 2025).
d. Triple Extraction	Prompt LLM (<code>gemma2:9b</code> or <code>mistral</code>) with contextual windows to extract MALONT-formatted triples (subject, predicate, object, evidence).
e. Normalization and Mapping	Normalize MALONT types & predicates → map to Connection ontology via <code>malont_adapter.normalize_and_map_malont</code> .
f. Ontology-Aware Validation	Validate mapped triples against Connection ontology rules in <code>validator.py</code> using <code>validate_triple_record()</code> .

g. Graph Insertion & Embedding Insert validated triples into Neo4j as :CTIEntity nodes and embed with nomic-embed-text for similarity analysis.

2. Workflow Overview

3. Challenges Observed

1. Chunk Granularity Tuning:

- Fixed window or threshold values (`min_sim_threshold` $\approx 0.35-0.45$) significantly affect chunk size and context coherence.
- Need dynamic thresholding based on in-domain similarity distributions.

2. Embedding Stability:

- Ollama embedding API occasionally drops connections or returns `None`; added vector normalization and `None` guards in `_ollama_embed`.

3. Parsing Noise from LLM Outputs:

- Mixed return formats (`list/dict/string`) caused too many values to unpack; resolved via robust `_coerce_triple_list()` parser with nested JSON handling.

4. Validation Leniency:

- Ontology mapping gaps (e.g., unlisted predicates like `investigates`) currently labeled as `lenient:unknown-predicate`; need to extend ontology relations for CTI domain coverage.

5. Reranking Need:

- Extracted triples can be ranked by cosine similarity to chunk centroids or semantic relevance to primary topic for final selection (preliminary idea for next week).

4. Next Steps / Planned Work

- Integrate Max–Min Chunking Module: finalize the look-ahead window (4–8) and soft size cap (≈ 200 tokens).
- Implement Chunk-Level Reranking: use semantic coherence scores to re-order triples before validation.
- Benchmark Validation: compare precision/recall against CTI-HAL gold annotations for 10–15 reports.
- Documentation & Paper Prep: draft methods and evaluation section for a conference submission by early December (6 weeks goal).